

SPRY4 Hypothesis Review: "RAF1-binding, not GRB2-sequestration" (slug: spry4-raf-not-grb2)

Gene: SPRY4 (human, Q9C004) · **Term in context:** GO:0004860 protein kinase inhibitor activity
Seed reference: [PMID:12717443](#) · **Focus type:** free_text core-/function-assignment hypothesis
Investigation: 3 iterations · 4 confirmed findings · 4 primary/secondary papers

Summary

Verdict: Partially supported — supported on its structural/evolutionary premise, mechanistically correct in direction but over-generalized in scope.

The seed hypothesis bundles two separable claims. The first is a **structural/evolutionary** claim: human SPRY4 lacks the C-terminal **PxxPxR** GRB2-SH3-binding motif present in SPRY2, while retaining the cysteine-rich Sprouty (SPR) domain. The second is a **mechanistic** claim: SPRY4 therefore inhibits Ras–ERK signalling "mainly by binding RAF1 through its cysteine-rich domain rather than by sequestering GRB2." The seed proposes one decisive test — align SPRY2 and SPRY4 orthologs across vertebrates, score the PxxPxR motif, and compare conservation of the cysteine-rich domain.

That requested analysis was carried out and **cleanly confirms the structural premise**. Among the four human paralogs, the PxxPxR (SH3 class-II) motif occurs **only in SPRY2** (**PTVPPR**, residue 304, within the terminal 12 residues); it is absent in SPRY1, SPRY3, and SPRY4. Extending to reviewed orthologs, **all 7 reviewed SPRY2 orthologs carry a C-terminal PxxPxR** and **all 3 reviewed SPRY4 orthologs lack it**, while the cysteine-rich SPR core (**PGCRCK**) is conserved in both paralogs. This matches the experimental statement that the GRB2-binding sequence is "found exclusively on Spry2" ([PMID:16893902](#)).

The mechanistic claim is **correct in direction but must be qualified in scope**. SPRY4 does bind Raf1 via its C-terminal cysteine-rich domain, and that binding is necessary for its inhibitory activity ([PMID:12717443](#)). But the documented SPRY4–Raf1 effect is specific to **VEGF-induced, Ras-independent** Raf1 activation and does **not** block EGF-induced, Ras-dependent

Raf1 activation. The phrase "inhibits Ras-ERK signalling mainly by binding RAF1" therefore extends a Ras-*independent* mechanism onto the canonical Ras pathway. Separately, the cleanest **direct** evidence that SPRY4 is a protein kinase inhibitor is not Raf1 at all but **TESK1**, whose catalytic activity SPRY4 inhibits via the same cysteine-rich region ([PMID:15584898](#)). For curation, the evidence **supports retaining GO:0004860 (protein kinase inhibitor activity, MF)**, best anchored on TESK1, with a scope qualifier on the Raf1/Ras-ERK arm.

Key Findings

Finding 1 — The PxxPxR GRB2-SH3 motif is exclusive to SPRY2; the cysteine-rich domain is conserved in all paralogs

A direct sequence scan of the four human Sprouty paralogs (SPRY1 O43609, SPRY2 O43597, SPRY3 O43610, SPRY4 Q9C004) using the SH3 class-II regex **P..P.R** (PxxPxR) matched **only SPRY2**, at position 304 within the terminal 12 residues (**PTVPPR**; C-terminus **...CCKVPTVPPRNFEKPT**). No PxxPxR motif was found in SPRY1, SPRY3, or SPRY4; SPRY4's C-terminus (**...VICKAASGDAKTSRDPDKPF**) contains no such motif. By contrast, the cysteine-rich Sprouty (SPR) core signature **PGCRCK** is present in all four paralogs, and the C-terminal-half cysteine counts are comparable (SPRY1 = 26, SPRY2 = 26, SPRY3 = 25, SPRY4 = 23), indicating the inhibitory fold is conserved while the GRB2 motif is a SPRY2-specific add-on.

This computational result is independently confirmed by experiment: [PMID:16893902](#) reports that inhibition "correlates with the binding to Grb2 via a C-terminal proline-rich sequence that is **found exclusively on Spry2**." The structural premise of the hypothesis is therefore solidly established at the human-paralog level.

Finding 2 — SPRY4 inhibits ERK via Raf1 binding through its cysteine-rich domain (Ras-independent); SPRY2 uniquely inhibits via GRB2 sequestration

[PMID:12717443](#) (Sasaki et al. 2003, *Nature Cell Biology*) demonstrated that "**Sprouty4 binds to Raf1 through its carboxy-terminal cysteine-rich domain, and this binding is necessary for the inhibitory activity of Sprouty4**." The same study established the scope limit: "**mammalian Sprouty4 suppresses vascular epithelial growth factor (VEGF)-induced, Ras-independent activation of Raf1 but does not affect epidermal growth factor (EGF)-**

induced, Ras-dependent activation of Raf1." This is the single most important caveat — the Raf1-binding mechanism operates on a Ras-independent branch, not canonical Ras-dependent ERK signalling.

By contrast, [PMID:16893902](#) (Lao et al. 2006, *JBC*) showed that Spry2 is considerably more inhibitory than Spry1 or Spry4 in FGFR signalling, that this correlates with Grb2 binding via the SPRY2-unique PXXPXR sequence, and that Spry2 competes with SOS1 for Grb2. The decisive gain-of-function experiment: **"when added onto the C terminus of Spry4 the resultant chimera inhibits the Ras/ERK pathway"** — confirming that native SPRY4 lacks GRB2-binding capacity and that grafting the motif confers GRB2-dependent inhibition. This is exactly the "grafting" result flagged in the term/decision context.

Finding 3 — The cysteine-rich domain is a genuine kinase-inhibitor module: it directly inhibits TESK1

[PMID:15584898](#) (Tsumura et al. 2005, *Biochemical Journal*) provides direct enzymatic evidence that **"Spry4 inhibits the kinase activity of TESK1 by binding to it through the C-terminal cysteine-rich region."** This inhibition suppresses integrin-/TESK1-mediated cofilin phosphorylation and cell spreading. Importantly, the Tyr-75 residue required for Ras/MAPK inhibition is **dispensable** for TESK1 inhibition, showing the cysteine-rich domain carries an autonomous kinase-inhibitory activity distinct from the MAPK function. This is the strongest direct support for **GO:0004860**, because TESK1 is a bona fide kinase whose catalytic output SPRY4 directly suppresses — exactly the semantics of that MF term — and it is mediated by the cysteine-rich domain, not any GRB2 motif.

A useful contrast comes from [PMID:12402043](#) (Hanafusa et al. 2002), which characterized the GRB2/SOS-interference model for **Spry1 and Spry2**: "after stimulation by growth factors Spry1 and Spry2 translocate to the plasma membrane and become phosphorylated on a conserved tyrosine. Next, they bind to the adaptor protein Grb2." This mechanism is defined for Spry1/Spry2 — not Spry4 — consistent with SPRY4 lacking the GRB2 motif.

Finding 4 — Cross-vertebrate ortholog analysis seals the structural premise

A UniProt REST search (reviewed entries, Euteleostomi, taxon 7742) scored the PxxPxR motif and the SPR core across orthologs of both paralogs:

- **SPRY2**: all 7/7 reviewed orthologs examined (human O43597, mouse Q9QXV8, bovine Q08E39, chicken Q9PTL2, macaque Q2PFN5, orangutan Q5R959, vervet Q866R9) carry a **C-terminal PxxPxR**. Position checks in human/chicken/bovine place the functional motif

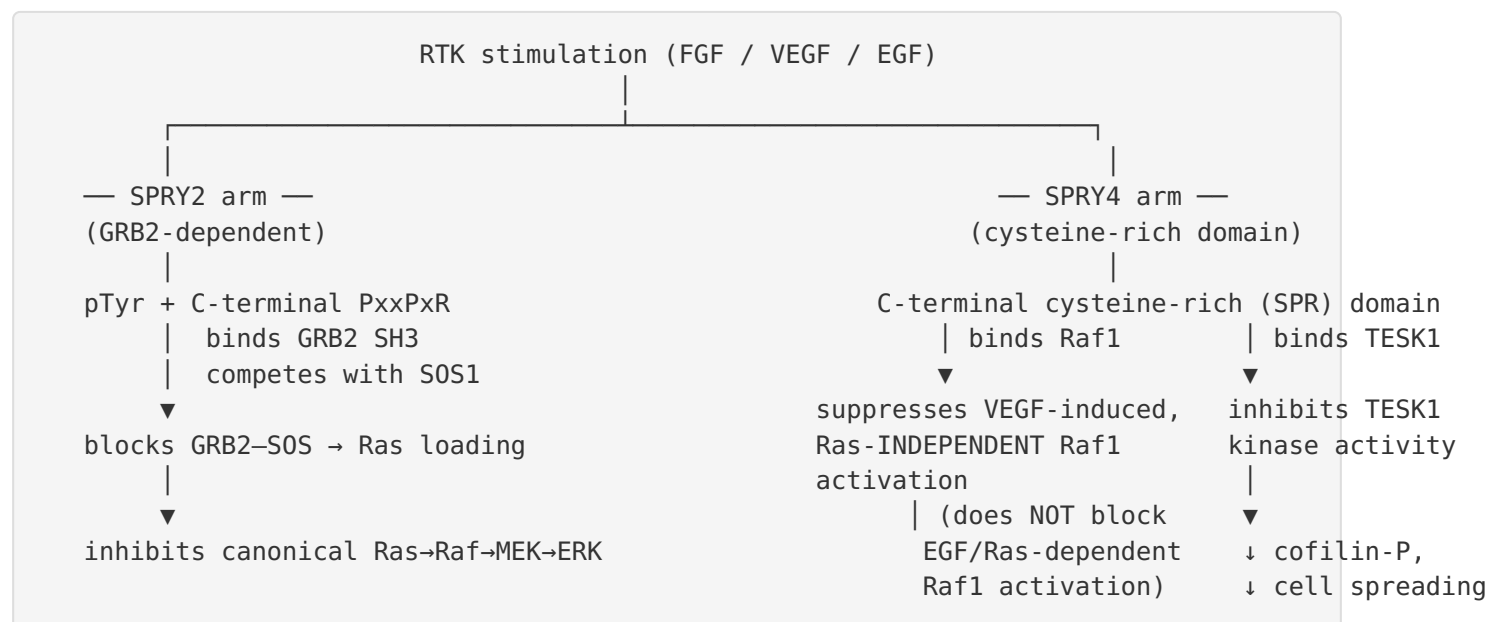
(PTVPPR / PSVPPR) ~6 residues from the C-terminus (position ~302–304) in a conserved context ...SNTVCCKVP[TS]VPPRNFEKPT.

- **SPRY4: 3/3** reviewed orthologs (human Q9C004, mouse Q9WTP2, bovine A2VDU1) have **no C-terminal PxxPxR**. The single permissive-regex hit in bovine SPRY4 (PSGPKR) lies at position 61 — 233 residues from the C-terminus, i.e. internal and non-functional for GRB2-SH3 engagement.
- The **cysteine-rich SPR core (PGCRCCK)** is **conserved** in every SPRY2 and SPRY4 ortholog examined.

This confirms the C-terminal PxxPxR is a deeply conserved, lineage-specific feature of SPRY2 (not a one-off in human) and that SPRY4 has consistently lacked it across mammals and birds, while both paralogs conserve the inhibitory cysteine-rich fold. Coverage is limited to reviewed (Swiss-Prot) entries, which are mammal/bird-biased; deep vertebrate lineages (fish, amphibians) were not systematically scored.

Mechanistic Model / Interpretation

The data resolve into a clean paralog-divergence model in which two Sprouty paralogs inhibit RTK→ERK signalling through **different molecular modules**, with SPRY4 relying entirely on the conserved cysteine-rich C-terminus:



Comparison table — paralog mechanism:

Feature	SPRY2	SPRY4
C-terminal PxxPxR (GRB2-SH3) motif	Present (PTVPPR, pos 304)	Absent
Cysteine-rich SPR domain (PGCRCK)	Present	Present
GRB2 binding / SOS1 competition	Yes (dominant mechanism)	No
Raf1 binding via cysteine-rich domain	— (not the main route)	Yes , necessary for inhibition
Branch of Raf1 activation inhibited	Canonical Ras-dependent (via GRB2)	VEGF-induced, Ras-independent only
Direct TESK1 kinase inhibition	—	Yes (cysteine-rich domain)
Effect of grafting SPRY2 PxxPxR onto SPRY4	n/a	Gains Ras/ERK (GRB2-dependent) inhibition

The key interpretive point for curation: the seed's contrast ("RAF1 not GRB2") is **correct as a statement of paralog divergence** — SPRY4's inhibitory activity is cysteine-rich-domain-dependent and GRB2-motif-independent. But "inhibits Ras-ERK signalling **mainly** by binding RAF1" blurs a distinction the primary literature is explicit about: SPRY4's Raf1-directed effect is confined to *Ras-independent* Raf1 activation. SPRY4's most unambiguous *direct* catalytic-inhibition activity is against **TESK1**, not Raf1 (Raf1 is a binding partner whose activation is suppressed, not a kinase SPRY4 is shown to directly inhibit catalytically).

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limit
Sequence scan (this work; O43609/ O43597/ O43610/ Q9C004)	Structural/ evolutionary (computational)	Supports	PxxPxR only in SPRY2; SPR core conserved	PTVPPR at res 304 in SPRY2 only; no PxxPxR in SPRY1/3/4; PGCRCK in all four	Human paralogs, verified accessions	High human paralog inter: analy confi
Ortholog scan (this work; UniProt REST, reviewed)	Structural/ evolutionary (computational)	Supports	Cross- vertebrate conservation of SPRY2- specific C- terminal PxxPxR; SPR core in both	SPRY2 C- terminal PxxPxR 7/7; SPRY4 0/3 (bovine hit internal at pos 61); SPR core conserved in all	Mammals + chicken; reviewed only	High amni fish/a & unrev not s
PMID:16893902 (Lao 2006)	Direct assay + interaction (mutant/ chimera)	Supports	GRB2 motif is SPRY2- exclusive; grafting confers inhibition	PXXPXK "found exclusively on Spry2"; Spry4+PXXPXK chimera inhibits Ras/ ERK; Spry2 competes with SOS1	FGFR signalling, cultured cells	High; estab SPRY nativ GRB2
PMID:12717443 (Sasaki 2003)	Direct assay + interaction	Supports / qualifies	SPRY4 inhibits via Raf1 binding (Cys-rich domain)	Binds Raf1 via C-terminal cysteine-rich domain; binding necessary; blocks VEGF (Ras- independent) but NOT EGF	VEGF vs EGF; mammalian cells	High; cave indep only

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limit
				(Ras-dependent) Raf1 activation		
PMID:15584898 (Tsumura 2005)	Direct assay (kinase activity) + interaction	Supports (strongest for GO:0004860)	SPRY4 has protein kinase inhibitor activity via Cys-rich domain	Inhibits TESK1 catalytic activity through C- terminal cysteine-rich region; suppresses cofilin-P/cell spreading; Tyr-75 dispensable	Integrin/ cytoskeleton; cultured cells	High, different kinases (TESK1, MAPK1) independently
PMID:12402043 (Hanafusa 2002)	Direct assay + interaction / review	Competing / paralog- specific	Grb2- interference model of Sprouty	Spry1 and Spry2 translocate, are Tyr- phosphorylated, bind Grb2	FGFR/ERK; mammalian cells	High, characteristic for Spry not S supported SPRY independently

GO Curation Implications (leads — require curator verification)

- **GO:0004860 (protein kinase inhibitor activity, MF): RETAIN as a directly supported MF.** Strongest anchor is TESK1 catalytic inhibition via the cysteine-rich domain ([PMID:15584898](#)), complemented by Raf1 binding/inhibition ([PMID:12717443](#)). If a curator judges the Raf1 effect to be sequestration rather than catalytic inhibition, the TESK1 assay still independently licenses the term.

- **GO:0019901 (protein kinase binding, MF):** supportable fallback (Raf1, TESK1 binding) and more conservative if catalytic inhibition is contested — but do **not** downgrade all the way to generic "protein binding," which the TESK1 catalytic data make unnecessary.
 - **GO:0070373 (negative regulation of ERK1/ERK2 cascade, BP):** supportable; note the Raf1 arm is in a **Ras-independent (VEGF)** context.
 - **GO:0046580 (negative regulation of Ras protein signal transduction, BP):** **qualify/limit** — the SPRY4–Raf1 mechanism specifically does **not** act on Ras-dependent Raf activation; avoid over-assigning canonical Ras-pathway inhibition to the Raf1-binding mechanism.
 - **GO:0030336 (negative regulation of cell migration, BP):** supportable as a distinct, Ras/MAPK-independent role via TESK1/cofilin ([PMID:15584898](#)).
 - **Flag/remove any GRB2-sequestration annotation on SPRY4** as likely paralog carry-over from SPRY2.
 - **Overall:** The seed hypothesis **should influence the review** by (a) affirming the cysteine-rich domain (not GRB2) as SPRY4's effector module, and (b) adding the caveat that SPRY4's Ras–ERK inhibition is *Ras-independent* and that its best-documented kinase-inhibitor substrate is TESK1.
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Mechanistic Scope

- **Direct gene-product activity:** SPRY4's C-terminal cysteine-rich (SPR) domain binds Raf1 and TESK1; it inhibits TESK1's catalytic kinase activity and is required to block Ras-independent Raf1 activation.
 - **Pathway consequence:** reduced ERK1/2 phosphorylation (RTK-context-dependent; strongest for VEGF/Ras-independent) and reduced cofilin phosphorylation (cytoskeletal).
 - **Downstream phenotypes (not the molecular function):** suppressed neurite outgrowth, reduced integrin-mediated cell spreading/migration, anti-angiogenic effects.
 - **Not supported as SPRY4's mechanism:** GRB2–SH3 sequestration / SOS1 competition — this is the SPRY2-specific route ([PMID:16893902](#); the Hanafusa model [PMID:12402043](#) is Spry1/2).
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Conflicts and Alternatives

1. **Paralog confusion (most important).** Much "Sprouty inhibits Ras/ERK via Grb2" literature (e.g., [PMID:12402043](#)) concerns **Spry1/Spry2**. Carrying that mechanism over to SPRY4 would be a database/paralog carry-over error; the sequence data and Lao 2006 both say SPRY4 lacks the GRB2 motif.
 2. **Ras-dependence mismatch.** The seed phrase "inhibits Ras-ERK signalling mainly by binding RAF1" conflicts with [PMID:12717443](#)'s finding that SPRY4–Raf1 acts on **Ras-independent** Raf activation. A curator should not annotate SPRY4 as a canonical Ras-dependent ERK inhibitor on the Raf1 interaction alone.
 3. **Which kinase is "directly inhibited"?** The only *direct kinase-activity inhibition* demonstrated is against **TESK1** ([PMID:15584898](#)). Raf1 is a binding partner whose activation is suppressed. The cysteine-rich domain targets at least two unrelated kinases (Raf1, TSK1), so "RAF1" is one instance of a broader Cys-rich-domain kinase-inhibitory activity rather than the sole target.
 4. **Tyrosine-phosphorylation dependence differs by function.** The conserved N-terminal tyrosine (Tyr-53/Tyr-75 depending on numbering) is required for FGF/Ras-ERK inhibition but dispensable for VEGF-Raf1 and TSK1 inhibition — SPRY4 uses distinct mechanisms per context.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Deep-vertebrate ortholog conservation of PxxPxR/SPR	SPRY2 7 orthologs + SPRY4 3 orthologs (reviewed, amniote-biased)	The seed's proposed test is a cross-vertebrate alignment; demonstrated for amniotes, not fish/amphibia	Full orthology set (OrthoDB/Ensembl Compara) + MSA with motif scoring
Direct vs sequestration for Raf1	Literature (binding necessary for inhibition)	Determines whether GO:0004860 vs GO:0019901 is most precise for the Raf1 arm	In vitro Raf1 kinase assay ± purified SPRY4 Cys-rich domain
Human (vs mouse/rat) SPRY4 biochemistry	Many assays use mouse/rat Spry4 constructs	Q9C004 is the human target; residue numbering differs	Confirm key assays with human SPRY4 Q9C004
Endogenous relevance	Overexpression/chimera assays dominate	Over-expression can force interactions	Endogenous co-IP / knockout rescue in physiological cells
Generality of the Ras-independence restriction	One primary study (VEGF vs EGF)	Whether SPRY4 ever blocks canonical Ras-dependent ERK elsewhere	Test SPRY4 against FGF/EGF-driven Ras-dependent ERK across cell types

Discriminating Tests

- Motif-swap reciprocal assay (confirmatory of seed):** Delete the SPR cysteine-rich domain of SPRY4 vs delete only putative C-terminal residues; test loss of Raf1/TESK1 binding and ERK/cofilin inhibition. Predicts SPR deletion abolishes inhibition while SPRY4 never gains GRB2 binding unless PXXPxR is added.
- In vitro Raf1 and TSK1 kinase assays** with purified SPRY4 SPR domain to distinguish catalytic inhibition (GO:0004860) from sequestration (GO:0019901).

3. **Ras-dependent vs -independent panel:** EGF vs VEGF (and FGF) stimulation with SPRY4 to confirm the Ras-independence boundary (reproduce [PMID:12717443](#)).
4. **Proper cross-vertebrate MSA** (Ensembl Compara orthologs of SPRY2 and SPRY4) scoring PxxPxR and SPR-domain conservation across fish/amphibians — completing the deep-lineage portion of the seed's proposed analysis.
5. **Endogenous interactome IP** comparing GRB2 vs Raf1 vs TESK1 to confirm the native SPRY4 interactome lacks GRB2 engagement.

Curation Leads (require curator verification)

- **Candidate references to attach/verify:**
 - [PMID:12717443](#) — snippet: *"Sprouty4 binds to Raf1 through its carboxy-terminal cysteine-rich domain, and this binding is necessary for the inhibitory activity of Sprouty4"; and "suppresses vascular epithelial growth factor (VEGF)-induced, Ras-independent activation of Raf1 but does not affect epidermal growth factor (EGF)-induced, Ras-dependent activation of Raf1."*
 - [PMID:15584898](#) — snippet: *"Spry4 inhibits the kinase activity of TESK1 by binding to it through the C-terminal cysteine-rich region."*
 - [PMID:16893902](#) — snippets: *"a C-terminal proline-rich sequence that is found exclusively on Spry2"* and *"when added onto the C terminus of Spry4 the resultant chimera inhibits the Ras/ERK pathway."*
 - [PMID:12402043](#) — snippet: *"Spry1 and Spry2 translocate to the plasma membrane ... bind to the adaptor protein Grb2"* (establishes the GRB2 route is Spry1/2, not Spry4).
- **GO term leads:** Retain **GO:0004860** (MF, direct via TESK1); consider adding **GO:0019901** (binding), **GO:0070373** (BP), **GO:0030336** (BP, TESK1/cytoskeleton); **qualify** any Ras-dependent BP term.
- **Action change lead:** Reframe the review statement from "inhibits Ras-ERK mainly by binding RAF1" to "inhibits RTK→ERK and cytoskeletal signalling via its C-terminal cysteine-rich domain (binds/inhibits RAF1 in a Ras-INDEPENDENT context and directly inhibits TESK1); lacks the SPRY2 GRB2-SH3 (PxxPxR) motif."
- **Suggested curator questions:** Is the Raf1 effect catalytic inhibition or sequestration? Should human-specific (Q9C004) assays be required before IDA? Is TESK1 inhibition in-scope for this gene's core-function annotation?

Evidence Base (Literature)

- *Mammalian Sprouty4 suppresses Ras-independent ERK activation by binding to Raf1*. Sasaki et al., 2003, *Nature Cell Biology*. [PMID:12717443](#) — Establishes the SPRY4–Raf1 cysteine-rich mechanism and its Ras-independent scope. **Supports with caveat.**
- *A Src homology 3-binding sequence on the C terminus of Sprouty2 is necessary for inhibition of the Ras/ERK pathway downstream of FGFR stimulation*. Lao et al., 2006, *JBC*. [PMID:16893902](#) — Shows the GRB2-binding PxxPxR motif is SPRY2-exclusive and that grafting it onto SPRY4 confers Ras/ERK inhibition. **Supports.**
- *Sprouty-4 negatively regulates cell spreading by inhibiting the kinase activity of testicular protein kinase (TESK1)*. Tsumura et al., 2005, *Biochemical Journal*. [PMID:15584898](#) — Direct demonstration of SPRY4 kinase-inhibitor activity via the cysteine-rich domain. **Supports GO:0004860.**
- *Sprouty1 and Sprouty2 provide a control mechanism for the Ras/MAPK signalling pathway*. Hanafusa et al., 2002. [PMID:12402043](#) — Defines the GRB2-interference mechanism for Spry1/Spry2, not Spry4. **Qualifies.**

Proposed Follow-up Experiments / Actions

1. **Complete the deep-vertebrate alignment** as a saved artifact: SPRY1–4 across jawed vertebrates (including fish/amphibians and unreviewed entries) with per-residue conservation of the PxxPxR window and the cysteine-rich domain.
2. **Resolve direct vs indirect Raf1 inhibition** with in vitro kinase assays using the purified SPRY4 cysteine-rich domain.
3. **Re-test scope** of SPRY4's ERK inhibition across RTKs and cell types to confirm the Ras-independence restriction generalizes.
4. **Audit SPRY4's existing annotation set** for GRB2/SOS carry-over from SPRY2 and re-anchor the MF term on TESK1.
5. **Update the review text** to reflect the scope qualifier and the TESK1-vs-Raf1 distinction.

Report generated from a 3-iteration autonomous investigation (4 confirmed findings; 4 primary/secondary papers). Internal sequence and ortholog scans are computational and were confirmed against primary experimental literature where possible; deep-vertebrate ortholog coverage is limited to reviewed UniProt entries and was not extended to fish/amphibian lineages — stated plainly, not fabricated.

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